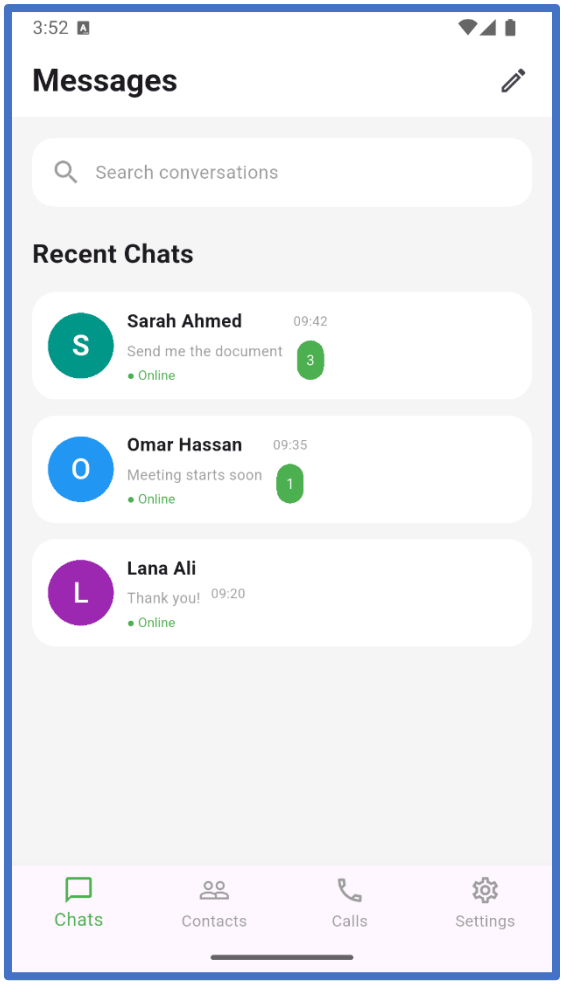
Practical 4

Example1: Design the following mobile application?




```
 import 'package:flutter/material.dart';


void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        backgroundColor: Colors.grey.shade100,
        appBar: AppBar(
          backgroundColor: Colors.white,
          title: const Text(
            'Messages',
            style: TextStyle(fontSize: 24,
fontWeight: FontWeight.bold),
          ),
          actions: const [Icon(Icons.edit_outlined), SizedBox(width:
18)],
        ),
        body: Padding(
          padding: const EdgeInsets.all(16),
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              Container(
                padding: const EdgeInsets.all(14),
                decoration: BoxDecoration(
                  color: Colors.white,
                  borderRadius: BorderRadius.circular(16),
                ),
                child: const Row(
                  children: [
                    Icon(Icons.search, color: Colors.grey),
                    SizedBox(width: 10),
                    Text(
                      'Search conversations',
                      style: TextStyle(color: Colors.grey),
                    ),
                  ],
                ),
```

```
              ),
              const SizedBox(height: 22),
              const Text(
                'Recent Chats',
                style: TextStyle(fontSize: 20, fontWeight:
FontWeight.bold),
              ),
              const SizedBox(height: 14),

              // Contact 1
              Container(
                margin: const EdgeInsets.only(bottom: 12),
                padding: const EdgeInsets.all(12),
                decoration: BoxDecoration(
                  color: Colors.white,
                  borderRadius: BorderRadius.circular(18),
                ),
                child: Row(
                  children: [
                    const CircleAvatar(
                      radius: 25,
                      backgroundColor: Colors.teal,
                      child: Text(
                        'S',
                        style: TextStyle(
                          color: Colors.white,
                          fontSize: 22,
                          fontWeight: FontWeight.bold,
                        ),
                      ),
                    ),
                    const SizedBox(width: 10),
                    Column(
                      crossAxisAlignment: CrossAxisAlignment.start,
                      children: [
                        Text(
                          'Sarah Ahmed',
                          style: TextStyle(
                            fontWeight: FontWeight.bold,
                            fontSize: 14,
                          ),
                        ),
                        SizedBox(height: 5),
                        Text(
                          'Send me the document',
                          style: TextStyle(color: Colors.grey, fontSize:
11),

                        ),
                        SizedBox(height: 5),
                        RichText(
```

```
                        text: TextSpan(
                          style: TextStyle(color: Colors.green,
fontSize: 10),
                          children: [
                            TextSpan(text: '● '),
                            TextSpan(text: 'Online'),
                          ],
                        ),
                      ),
                    ],
                  ),
                  const SizedBox(width: 8),
                  const Column(
                    children: [
                      Text(
                        '09:42',
                        style: TextStyle(color: Colors.grey, fontSize:
10),
                      ),
                      SizedBox(height: 8),
                      DecoratedBox(
                        decoration: BoxDecoration(
                          color: Colors.green,
                          borderRadius:
BorderRadius.all(Radius.circular(20)),
                        ),
                        child: Padding(
                          padding: EdgeInsets.all(7),
                          child: Text(
                            '3',
                            style: TextStyle(
                              color: Colors.white,
                              fontSize: 11,
                            ),
                          ),
                        ),
                      ),
                    ],
                  ),
                ],
              ),
            ),

            // Contact 2
            Container(
              margin: const EdgeInsets.only(bottom: 12),
              padding: const EdgeInsets.all(12),
              decoration: BoxDecoration(
                color: Colors.white,
                borderRadius: BorderRadius.circular(18),
```

```
                ),
                child: Row(
                  children: [
                    CircleAvatar(
                      radius: 25,
                      backgroundColor: Colors.blue,
                      child: Text(
                        'O',
                        style: TextStyle(color: Colors.white, fontSize:
22),
                      ),
                    ),
                    SizedBox(width: 10),
                    Column(
                      crossAxisAlignment: CrossAxisAlignment.start,
                      children: [
                        Text(
                          'Omar Hassan',
                          style: TextStyle(
                            fontWeight: FontWeight.bold,
                            fontSize: 14,
                          ),
                        ),
                        SizedBox(height: 5),
                        Text(
                          'Meeting starts soon',
                          style: TextStyle(color: Colors.grey, fontSize:
11),
                        ),
                        SizedBox(height: 5),
                        RichText(
                          text: TextSpan(
                            style: TextStyle(color: Colors.green,
fontSize: 10),
                            children: [
                              TextSpan(text: '● '),
                              TextSpan(text: 'Online'),
                            ],
                          ),
                        ),
                      ],
                    ),
                    SizedBox(width: 8),
                    Column(
                      children: [
                        Text(
                          '09:35',
                          style: TextStyle(color: Colors.grey, fontSize:
10),
                        ),
```

```
                        SizedBox(height: 8),
                        DecoratedBox(
                          decoration: BoxDecoration(
                            color: Colors.green,
                            borderRadius:
BorderRadius.all(Radius.circular(20)),
                          ),
                          child: Padding(
                            padding: EdgeInsets.all(7),
                            child: Text(
                              '1',
                              style: TextStyle(
                                color: Colors.white,
                                fontSize: 11,
                              ),
                            ),
                          ),
                        ),
                      ],
                    ),
                  ],
                ),
              ),

              // Contact 3
              Container(
                padding: const EdgeInsets.all(12),
                decoration: BoxDecoration(
                  color: Colors.white,
                  borderRadius: BorderRadius.circular(18),
                ),
                child: Row(
                  children: [
                    CircleAvatar(
                      radius: 25,
                      backgroundColor: Colors.purple,
                      child: Text(
                        'L',
                        style: TextStyle(color: Colors.white, fontSize:
22),

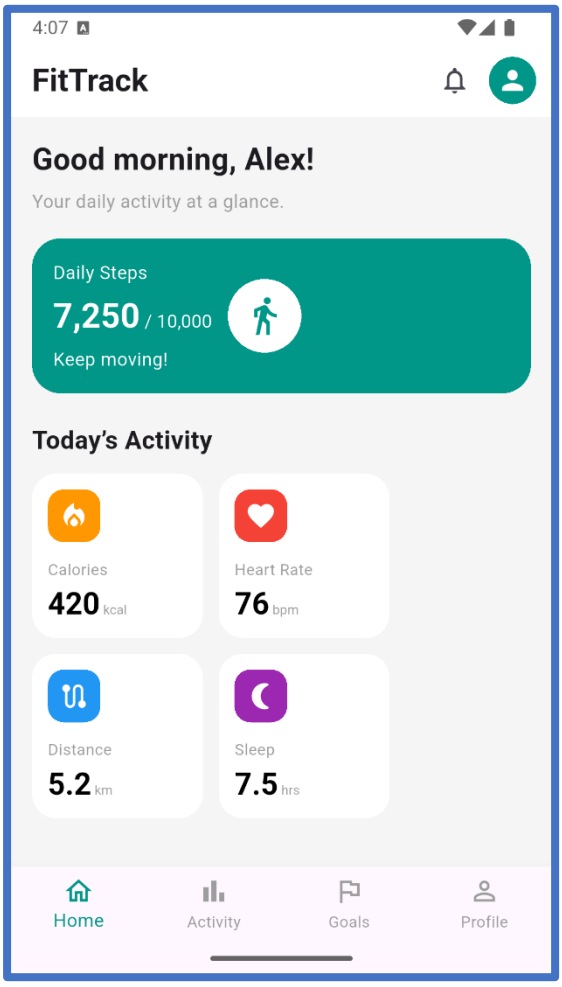
                      ),
                    ),
                    SizedBox(width: 10),
                    Column(
                      crossAxisAlignment: CrossAxisAlignment.start,
                      children: [
                        Text(
                          'Lana Ali',
                          style: TextStyle(
                            fontWeight: FontWeight.bold,
```

```
                            fontSize: 14,
                          ),
                        ),
                        SizedBox(height: 5),
                        Text(
                          'Thank you!',
                          style: TextStyle(color: Colors.grey, fontSize:
11),
                        ),
                        SizedBox(height: 5),
                        RichText(
                          text: TextSpan(
                            style: TextStyle(color: Colors.green,
fontSize: 10),
                            children: [
                              TextSpan(text: '● '),
                              TextSpan(text: 'Online'),
                            ],
                          ),
                        ),
                      ],
                    ),
                    SizedBox(width: 8),
                    Text(
                      '09:20',
                      style: TextStyle(color: Colors.grey, fontSize: 10),
                    ),
                  ],
                ),
              ),
            ],
          ),
        ),
        bottomNavigationBar: BottomNavigationBar(
          type: BottomNavigationBarType.fixed,
          currentIndex: 0,

          selectedItemColor: Colors.green,
          unselectedItemColor: Colors.grey,
          items: const [
            BottomNavigationBarItem(
              icon: Icon(Icons.chat_bubble_outline),
              label: 'Chats',
            ),
            BottomNavigationBarItem(
              icon: Icon(Icons.people_outline),
              label: 'Contacts',
            ),
            BottomNavigationBarItem(
              icon: Icon(Icons.call_outlined),
```

```
            label: 'Calls',
          ),
          BottomNavigationBarItem(
            icon: Icon(Icons.settings_outlined),
            label: 'Settings',
          ),
        ],
      ),
    ),
  );
 }
}
```

Example2: Write Flutter\ Dart code to design the following mobile application?




```
import 'package:flutter/material.dart';

void main() {
  runApp(const FitnessApp());
}

class FitnessApp extends StatelessWidget {
  const FitnessApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        backgroundColor:
Colors.grey.shade100,
        appBar: AppBar(
          backgroundColor: Colors.white,
          title: const Text(
            'FitTrack',
            style: TextStyle(fontSize: 24,
fontWeight: FontWeight.bold),
          ),
          actions: const [
            Icon(Icons.notifications_none),
```

```
            SizedBox(width: 14),
            CircleAvatar(
              radius: 18,
              backgroundColor: Colors.teal,
              child: Icon(Icons.person, color: Colors.white),
            ),
            SizedBox(width: 12),
          ],
        ),
        body: Padding(
          padding: const EdgeInsets.all(16),
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              const Text(
                'Good morning, Alex!',
                style: TextStyle(fontSize: 23, fontWeight:
FontWeight.bold),
              ),
              const SizedBox(height: 5),
              const Text(
                'Your daily activity at a glance.',
                style: TextStyle(color: Colors.grey),
              ),
              const SizedBox(height: 18),

              // Daily goal card
              Container(
                width: double.infinity,
                padding: const EdgeInsets.all(16),
                decoration: BoxDecoration(
                  color: Colors.teal,
                  borderRadius: BorderRadius.circular(22),
                ),
                child: Row(
                  children: [
                    Column(
                      crossAxisAlignment: CrossAxisAlignment.start,
                      children: [
                        Text(
                          'Daily Steps',
                          style: TextStyle(color: Colors.white, fontSize:
14),

                        ),
                        SizedBox(height: 8),
                        RichText(
                          text: TextSpan(
                            style: TextStyle(color: Colors.white),
                            children: [
                              TextSpan(
```

```
                            text: '7,250',
                            style: TextStyle(
                              fontSize: 26,
                              fontWeight: FontWeight.bold,
                            ),
                          ),
                          TextSpan(text: ' / 10,000'),
                        ],
                      ),
                    ),
                    SizedBox(height: 8),
                    Text(
                      'Keep moving!',
                      style: TextStyle(color: Colors.white),
                    ),
                  ],
                ),
                SizedBox(width: 12),
                CircleAvatar(
                  radius: 28,
                  backgroundColor: Colors.white,
                  child: Icon(
                    Icons.directions_walk,
                    size: 32,
                    color: Colors.teal,
                  ),
                ),
              ],
            ),
          ),

          const SizedBox(height: 22),
          const Text(
            'Today’s Activity',
            style: TextStyle(fontSize: 19, fontWeight:
FontWeight.bold),
          ),
          const SizedBox(height: 12),

          // Calories card
          Row(
            children: [
              Container(
                width: 130,
                padding: const EdgeInsets.all(12),
                decoration: BoxDecoration(
                  color: Colors.white,
                  borderRadius: BorderRadius.circular(18),
                ),
                child: Column(
```

```
                    crossAxisAlignment: CrossAxisAlignment.start,
                    children: [
                      DecoratedBox(
                        decoration: BoxDecoration(
                          color: Colors.orange,
                          borderRadius:
BorderRadius.all(Radius.circular(12)),
                        ),
                        child: Padding(
                          padding: EdgeInsets.all(8),
                          child: Icon(
                            Icons.local_fire_department,
                            color: Colors.white,
                          ),
                        ),
                      ),
                      SizedBox(height: 12),
                      Text(
                        'Calories',
                        style: TextStyle(color: Colors.grey, fontSize:
12),
                      ),
                      SizedBox(height: 5),
                      RichText(
                        text: TextSpan(
                          style: TextStyle(color: Colors.black),
                          children: [
                            TextSpan(
                              text: '420',
                              style: TextStyle(
                                fontSize: 23,
                                fontWeight: FontWeight.bold,
                              ),
                            ),
                            TextSpan(
                              text: ' kcal',
                              style: TextStyle(
                                color: Colors.grey,
                                fontSize: 10,
                              ),
                            ),
                          ],
                        ),
                      ),
                    ],
                  ),
                ),
                const SizedBox(width: 12),

                // Heart rate card
```

```
                  Container(
                    width: 130,
                    padding: const EdgeInsets.all(12),
                    decoration: BoxDecoration(
                      color: Colors.white,
                      borderRadius: BorderRadius.circular(18),
                    ),
                    child: Column(
                      crossAxisAlignment: CrossAxisAlignment.start,
                      children: [
                        DecoratedBox(
                          decoration: BoxDecoration(
                            color: Colors.red,
                            borderRadius:
BorderRadius.all(Radius.circular(12)),
                          ),
                          child: Padding(
                            padding: EdgeInsets.all(8),
                            child: Icon(Icons.favorite, color:
Colors.white),
                          ),
                        ),
                        SizedBox(height: 12),
                        Text(
                          'Heart Rate',
                          style: TextStyle(color: Colors.grey, fontSize:
12),
                        ),
                        SizedBox(height: 5),
                        RichText(
                          text: TextSpan(
                            style: TextStyle(color: Colors.black),
                            children: [
                              TextSpan(
                                text: '76',
                                style: TextStyle(
                                  fontSize: 23,
                                  fontWeight: FontWeight.bold,
                                ),
                              ),
                              TextSpan(
                                text: ' bpm',
                                style: TextStyle(
                                  color: Colors.grey,
                                  fontSize: 10,
                                ),
                              ),
                            ],
                          ),
                        ),
```

```
                    ],
                  ),
                ),
              ],
            ),

            const SizedBox(height: 12),

            // Distance card
            Row(
              children: [
                Container(
                  width: 130,
                  padding: const EdgeInsets.all(12),
                  decoration: BoxDecoration(
                    color: Colors.white,
                    borderRadius: BorderRadius.circular(18),
                  ),
                  child: Column(
                    crossAxisAlignment: CrossAxisAlignment.start,
                    children: [
                      DecoratedBox(
                        decoration: BoxDecoration(
                          color: Colors.blue,
                          borderRadius:
BorderRadius.all(Radius.circular(12)),
                        ),
                        child: Padding(
                          padding: EdgeInsets.all(8),
                          child: Icon(Icons.route, color:
Colors.white),
                        ),
                      ),
                      SizedBox(height: 12),
                      Text(
                        'Distance',
                        style: TextStyle(color: Colors.grey, fontSize:
12),
                      ),
                      SizedBox(height: 5),
                      RichText(
                        text: TextSpan(
                          style: TextStyle(color: Colors.black),
                          children: [
                            TextSpan(
                              text: '5.2',
                              style: TextStyle(
                                fontSize: 23,
                                fontWeight: FontWeight.bold,
                              ),
```

```
                                ),
                                TextSpan(
                                  text: ' km',
                                  style: TextStyle(
                                    color: Colors.grey,
                                    fontSize: 10,
                                  ),
                                ),
                              ],
                            ),
                          ),
                        ],
                      ),
                    ),
                    const SizedBox(width: 12),

                    // Sleep card
                    Container(
                      width: 130,
                      padding: const EdgeInsets.all(12),
                      decoration: BoxDecoration(
                        color: Colors.white,
                        borderRadius: BorderRadius.circular(18),
                      ),
                      child: Column(
                        crossAxisAlignment: CrossAxisAlignment.start,
                        children: [
                          DecoratedBox(
                            decoration: BoxDecoration(
                              color: Colors.purple,
                              borderRadius:
BorderRadius.all(Radius.circular(12)),
                            ),
                            child: Padding(
                              padding: EdgeInsets.all(8),
                              child: Icon(
                                Icons.nightlight_round,
                                color: Colors.white,
                              ),
                            ),
                          ),
                          SizedBox(height: 12),
                          Text(
                            'Sleep',
                            style: TextStyle(color: Colors.grey, fontSize:
12),
                          ),
                          SizedBox(height: 5),
                          RichText(
                            text: TextSpan(
```

```
                        style: TextStyle(color: Colors.black),
                        children: [
                          TextSpan(
                            text: '7.5',
                            style: TextStyle(
                              fontSize: 23,
                              fontWeight: FontWeight.bold,
                            ),
                          ),
                          TextSpan(
                            text: ' hrs',
                            style: TextStyle(
                              color: Colors.grey,
                              fontSize: 10,
                            ),
                          ),
                        ],
                      ),
                    ),
                  ],
                ),
              ),
            ),
          ],
        ),
      ],
    ),
  ),
  bottomNavigationBar: BottomNavigationBar(
    type: BottomNavigationBarType.fixed,
    currentIndex: 0,
    selectedItemColor: Colors.teal,
    unselectedItemColor: Colors.grey,
    items: const [
      BottomNavigationBarItem(
        icon: Icon(Icons.home_outlined),
        label: 'Home',
      ),
      BottomNavigationBarItem(
        icon: Icon(Icons.bar_chart),
        label: 'Activity',
      ),
      BottomNavigationBarItem(
        icon: Icon(Icons.flag_outlined),
        label: 'Goals',
      ),
      BottomNavigationBarItem(
        icon: Icon(Icons.person_outline),
        label: 'Profile',
      ),
    ], ),),);}}
```